# Predicting Falcon 9 First-Stage Landing Success

IBM Applied Data Science Capstone — SpaceX launch data, EDA, interactive analytics and machine learning

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October 31, 2026

https://github.com/YOUR-USERNAME/spacex-falcon9-capstone

# Outline

- Executive Summary
- Introduction
- Methodology
- Results
  - Insights drawn from EDA
  - Launch site proximity analysis (Folium)
  - Build a dashboard with Plotly Dash
  - Predictive analysis (classification)
- Conclusion
- Appendix

# Executive Summary

## Methods

- Data collection: SpaceX REST API (preserved IBM API snapshot) and Wikipedia web scraping with BeautifulSoup
- Data wrangling: outcome labels mapped to binary landing class; missing values handled
- EDA: Matplotlib visualizations and 12 SQL queries (SQLite)
- Interactive analytics: Folium launch-site maps and a Plotly Dash dashboard
- Prediction: Logistic Regression, SVM, Decision Tree and KNN tuned with GridSearchCV (5-fold CV)

## Key results

- **60/90** launches (66.7%) landed successfully, 2010–2020
- Landing success rose from **0% (2010–13)** to **90.0% (2019)**
- KSC LC-39A had the highest site success rate (**77.3%**); SSO, GEO, HEO, ES-L1 orbits 100%
- All four models scored **84.8%–88.9%** CV accuracy; **SVM** was best on CV
- Decision tree and KNN reached the highest test accuracy (**83.3%**)
- Main error: predicted landings that failed (false positives)

# Introduction

## Project background and context

- SpaceX advertises Falcon 9 launches at about $62 million; other providers charge upward of $165 million
- Much of the saving comes from recovering and reusing the first stage
- If we can predict whether the first stage will land, we can estimate the cost of a launch
- A competitor (Space Y) can use this to bid against SpaceX for launch contracts

### Problems we want to answer

- Which factors (payload mass, orbit, launch site, flight number, booster version) influence landing success?
- How have landing outcomes changed over time?
- Where are launch sites located, and what is near them?
- Can a machine learning model predict whether the first stage will land successfully?

Section 1

# Methodology

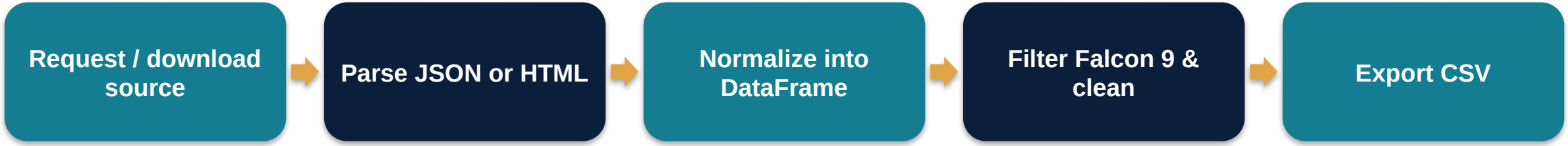
Data collection, wrangling, EDA, interactive visual analytics and predictive analysis

# Methodology — Executive Summary

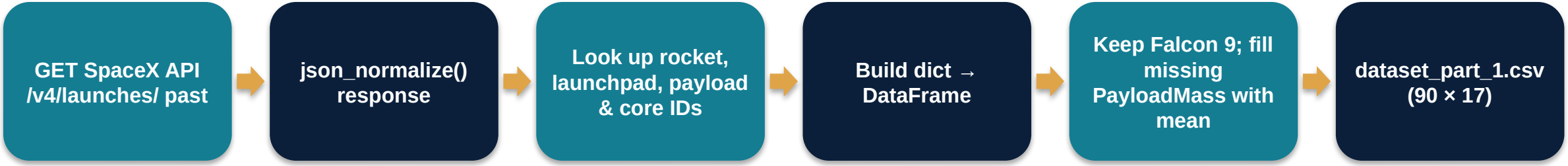
- **Data collection:** SpaceX REST API (launches, rockets, launch pads, payloads, cores) and web scraping of the Wikipedia 'List of Falcon 9 and Falcon Heavy launches' page
- **Data wrangling:** filtered to Falcon 9 launches; converted mission Outcome strings to a binary Class label (1 = landed, 0 = did not land); checked missing values
- **EDA with visualization and SQL:** scatter, bar and line charts; SQL queries on the historical launch table
- **Interactive visual analytics:** Folium maps (site markers, outcome markers, proximity lines) and a Plotly Dash dashboard
- **Predictive analysis:** scikit-learn pipelines (impute → scale → one-hot) with GridSearchCV over Logistic Regression, SVM, Decision Tree and KNN; evaluated on a stratified 20% hold-out test set

# Data Collection

- Two complementary sources were used to build the launch dataset:
  - SpaceX REST API — structured JSON records for each past launch (booster, payload, orbit, site, landing outcome)
  - Wikipedia — HTML tables of Falcon 9 launch records, parsed with BeautifulSoup
- Records were combined into a pandas DataFrame, filtered to Falcon 9, cleaned and saved to CSV for EDA and modeling



# Data Collection — SpaceX API



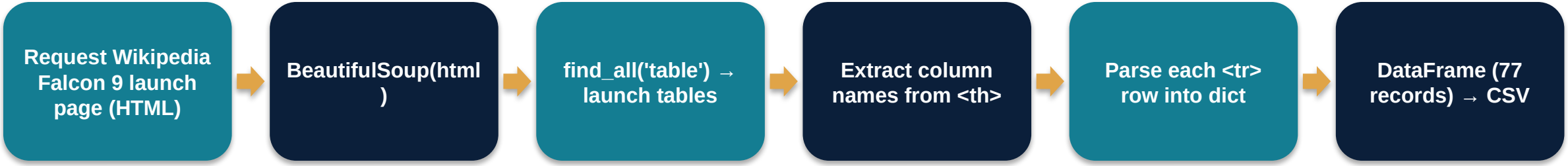

```
response = requests.get(
  'https://api.spacexdata.com/v4/launches/past')
data = pd.json_normalize(response.json())
getBoosterVersion(data); getLaunchSite(data)
getPayloadData(data);   getCoreData(data)
df = pd.DataFrame(launch_dict)
df = df[df.BoosterVersion != 'Falcon 1']
```

## Notes

- Live API returned HTTP 522 at run time
- Analysis uses the preserved IBM API snapshot (90 launches, 2010-06-04 to 2020-11-05)
- Full collection code: api_collection.py

GitHub: https://github.com/YOUR-USERNAME/spacex-falcon9-capstone/blob/main/api_collection.py (also: SpaceX_Capstone.ipynb)

# Data Collection — Web Scraping



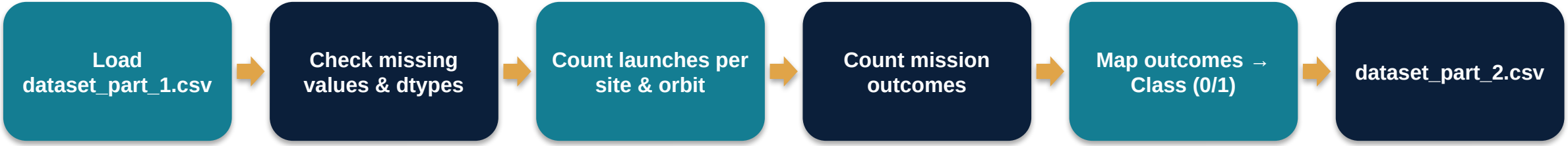

```
soup = BeautifulSoup(html, 'html.parser')
tables = soup.find_all('table', 'wikitable')
for row in table.find_all('tr'):
    cells = row.find_all('td')
    launch_dict['Booster'].append(...)
df = pd.DataFrame(launch_dict)
```

## Output

- 77 Falcon 9 launch records
- Fields: flight no., date/time, booster, site, payload, mass, orbit, customer, mission and landing outcome
- Parsed from saved Wikipedia HTML

GitHub: https://github.com/YOUR-USERNAME/spacex-falcon9-capstone/blob/main/visuals.py (also: SpaceX_Capstone.ipynb)

# Data Wrangling




- Outcome combines mission result and landing type, e.g. True ASDS, False Ocean, None None
- **Bad outcomes** → Class 0: False Ocean, False RTLS, False ASDS, None ASDS, None None
- All other outcomes (True ASDS, True RTLS, True Ocean) → Class 1
- Only LandingPad had missing values (26 rows, i.e. no pad used) — kept as-is
- Result: 60 Class 1 vs 30 Class 0 (success rate 66.7%)

```
bad = {'False Ocean','False RTLS',
       'False ASDS','None ASDS',
       'None None'}
df['Class'] = (~df.Outcome
   .isin(bad)).astype(int)
df.Class.mean()  # 0.667
```

GitHub: https://github.com/YOUR-USERNAME/spacex-falcon9-capstone/blob/main/analyze.py (also: SpaceX_Capstone.ipynb)

# EDA with Data Visualization

| Chart | Variables | Purpose |
|---|---|---|
| Scatter | Flight Number vs Launch Site | Does experience at a site improve landing? |
| Scatter | Payload Mass vs Launch Site | Do heavier payloads fail more at some sites? |
| Bar | Success rate by Orbit | Which orbit types have the highest success rate? |
| Scatter | Flight Number vs Orbit | Does the orbit trend change as flights accumulate? |
| Scatter | Payload Mass vs Orbit | How payload affects success for each orbit |
| Line | Yearly success rate | Trend of landing success over time |

Scatter plots show relationships between two variables coloured by outcome; bar charts compare categories; line charts show trends over time.

GitHub: https://github.com/YOUR-USERNAME/spacex-falcon9-capstone/blob/main/analyze.py (also: SpaceX_Capstone.ipynb)

# EDA with SQL

The historical launch table (101 rows) was loaded into SQLite and queried to:

- Display unique launch site names
- Show 5 records where site begins with 'CCA'
- Total payload carried for NASA (CRS)
- Average payload for booster F9 v1.1
- Date of first successful ground-pad landing
- Boosters with drone-ship success and 4,000–6,000 kg payload
- Count of successful and failed mission outcomes
- Boosters that carried the maximum payload (subquery)
- 2015 drone-ship failures with booster and site
- Rank landing outcomes between 2010-06-04 and 2017-03-20
- Landing success counts per launch site

GitHub: https://github.com/YOUR-USERNAME/spacex-falcon9-capstone/blob/main/queries.sql (also: analyze.py)

# Build an Interactive Map with Folium

- **Circles + markers** for each launch site (CCAFS LC-40, CCAFS SLC-40, KSC LC-39A, VAFB SLC-4E) to show where sites are
- **Marker clusters** of launch outcomes per site, green = success (Class 1), red = failure (Class 0), to see which sites perform best
- **MousePosition** to read coordinates of nearby features
- **PolyLines + distance markers** from CCAFS SLC-40 to the nearest coastline, road and railway, using the haversine distance
- Why: to check whether site location and surroundings (coast, logistics, safety distance from cities) relate to launch success

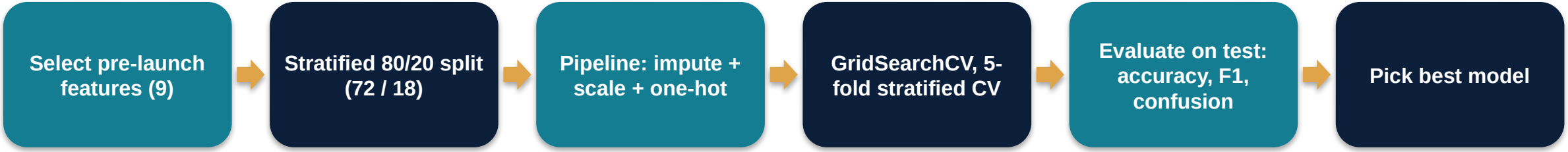
GitHub: https://github.com/YOUR-USERNAME/spacex-falcon9-capstone/blob/main/visuals.py (also: proximity.py)

# Build a Dashboard with Plotly Dash

- **Launch site dropdown** (All sites or one site) to filter the whole dashboard
- **Pie chart:** for All sites — share of successful launches by site; for one site — success vs failure counts
- **Payload range slider** (0–10,000 kg) to select a payload window
- **Scatter plot:** payload mass vs outcome class, coloured by booster version category
- Why: lets users interactively find which sites, payload ranges and booster versions have the highest success
- Data: 56-launch IBM dashboard snapshot (spacex_launch_dash.csv)

GitHub: https://github.com/YOUR-USERNAME/spacex-falcon9-capstone/blob/main/dashboard.py

# Predictive Analysis (Classification)




- Features: FlightNumber, PayloadMass, Flights, Block, GridFins, Reused, Legs, Orbit, LaunchSite
- Outcome-derived fields excluded to avoid leakage
- Preprocessing is fitted inside CV folds
- Extra check: train on first 72 flights, test on last 18 (time-ordered)

| Model | Grid searched |
|---|---|
| Logistic Regression | C |
| SVM | C, kernel, gamma |
| Decision Tree | max_depth, min_samples_leaf |
| KNN | n_neighbors, weights |

GitHub: https://github.com/YOUR-USERNAME/spacex-falcon9-capstone/blob/main/analyze.py (also: SpaceX_Capstone.ipynb)

# Results

### Exploratory analysis

- Success rate improves with flight number
- 2019 success rate 90.0%
- Orbit and site both matter

### Interactive analytics

- All sites near coast
- KSC LC-39A leads successful launches
- Payload 2–6 t has most successes

### Predictive analysis

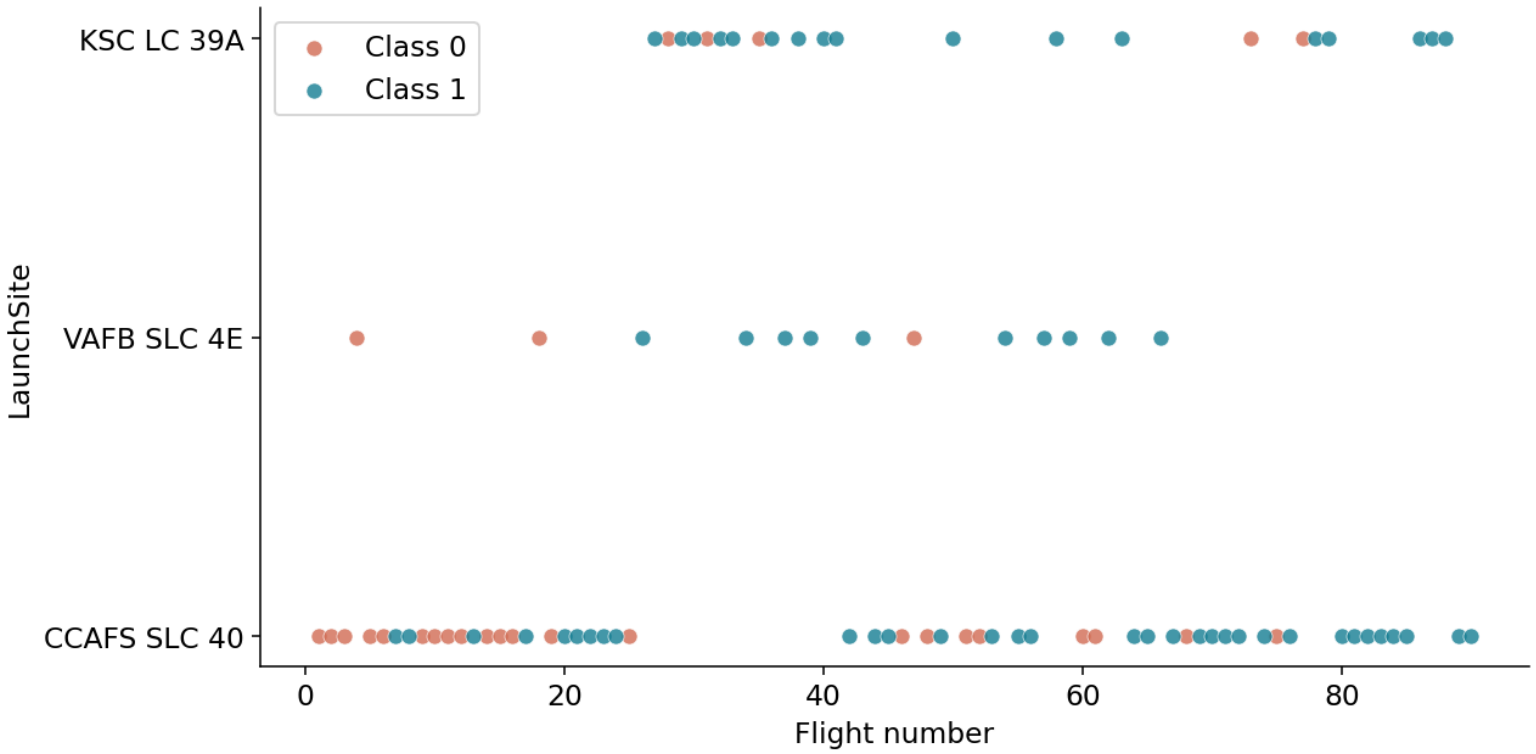
- Best CV: SVM 88.9%
- Best test: Tree/KNN 83.3%
- No false negatives on test

Section 2

# Insights Drawn from EDA

Visual and SQL exploration of the Falcon 9 launch data

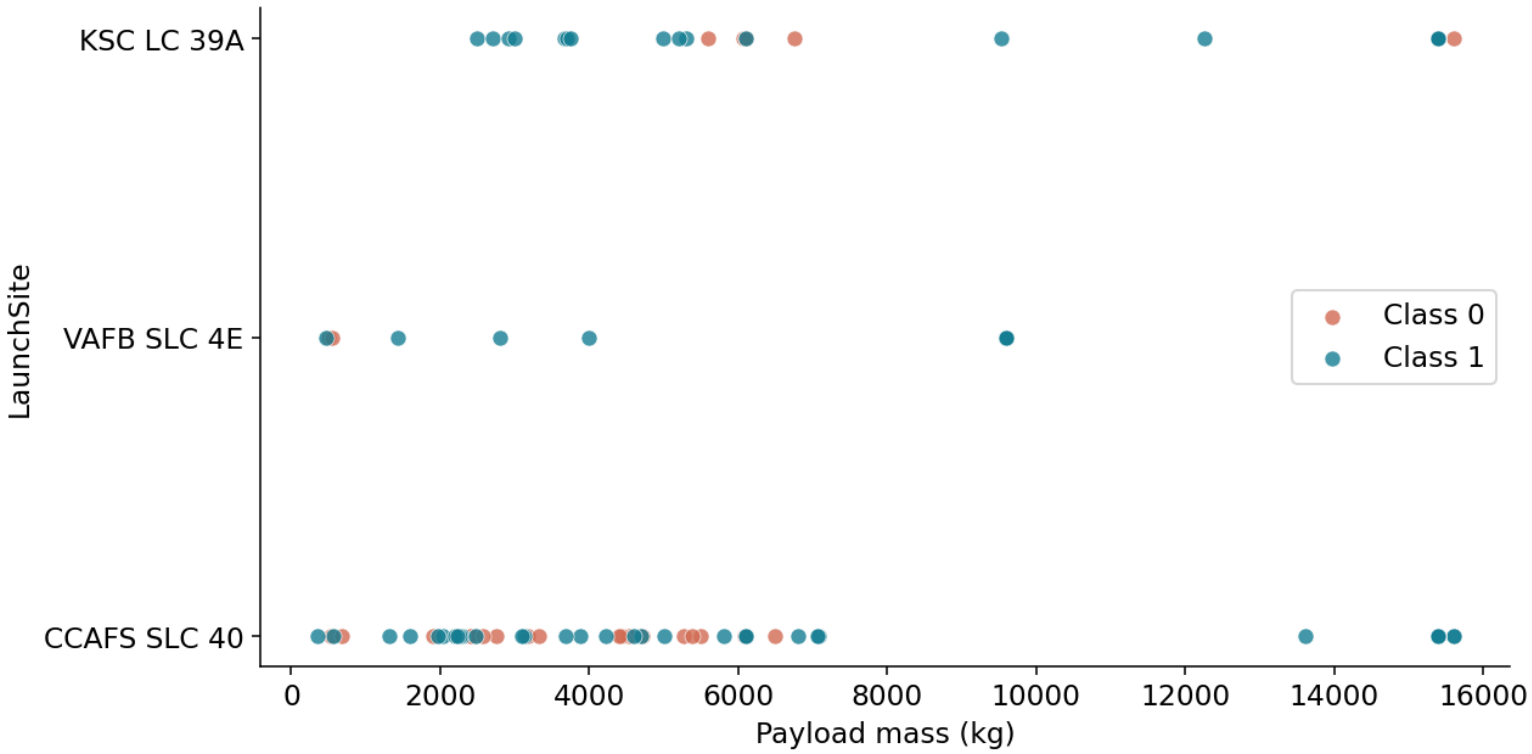
# Flight Number vs. Launch Site




## Insights

- Flights 1–25 (mostly CCAFS SLC 40) were largely failures
- After ~flight 40, most launches at every site succeed
- CCAFS SLC-40 has the most launches (55)
- More experience → higher landing success

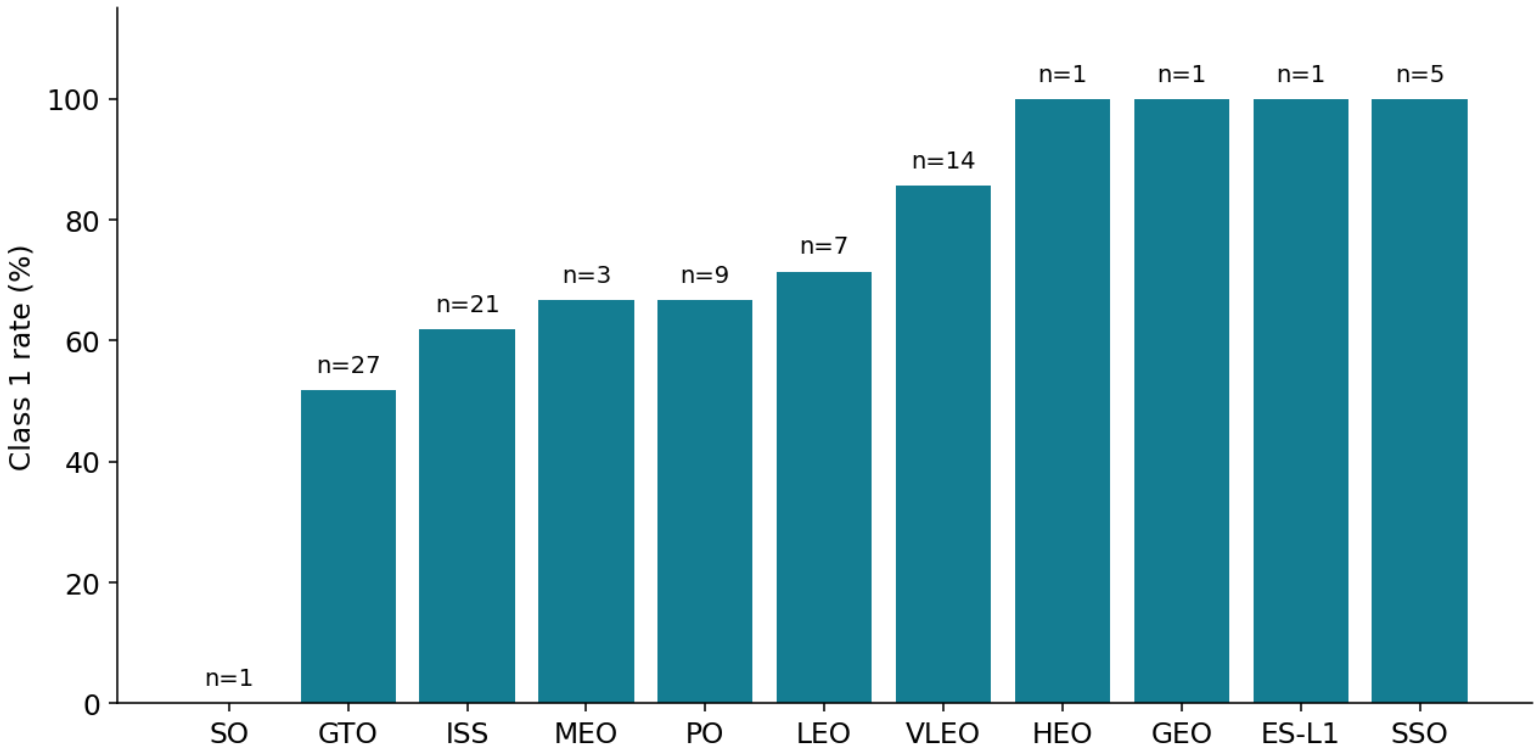
# Payload vs. Launch Site




## Insights

- VAFB SLC-4E launched no payloads above ~10,000 kg
- Heavy payloads (>8,000 kg) are mostly successful
- Light payloads at CCAFS show mixed outcomes
- Payload alone does not decide success

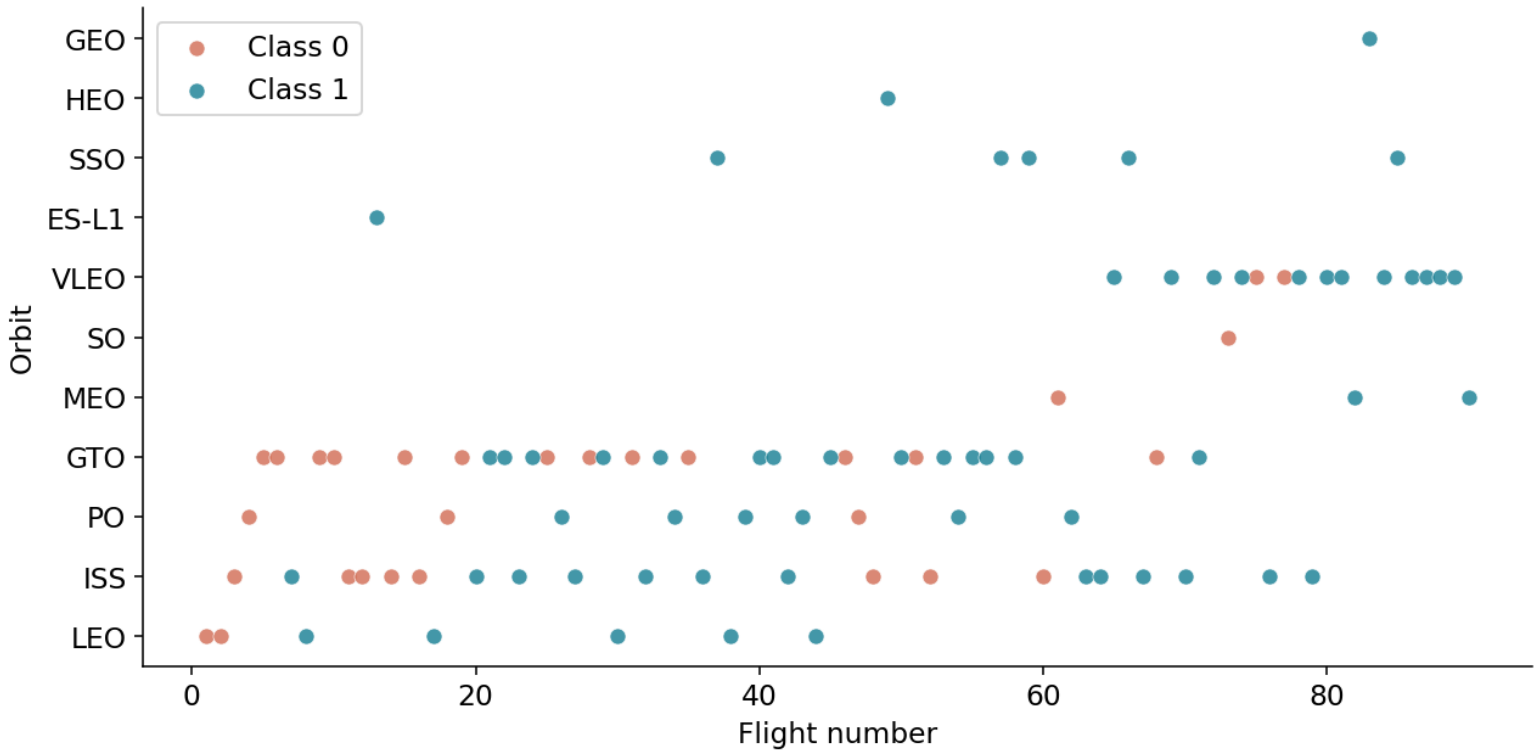
# Success Rate vs. Orbit Type




## Insights

- SSO (n=5), GEO, HEO and ES-L1 (n=1 each): 100%
- VLEO 85.7% on 14 launches — strongest large group
- GTO lowest of large groups at 51.9% (n=27)
- SO 0% (n=1); small groups need caution

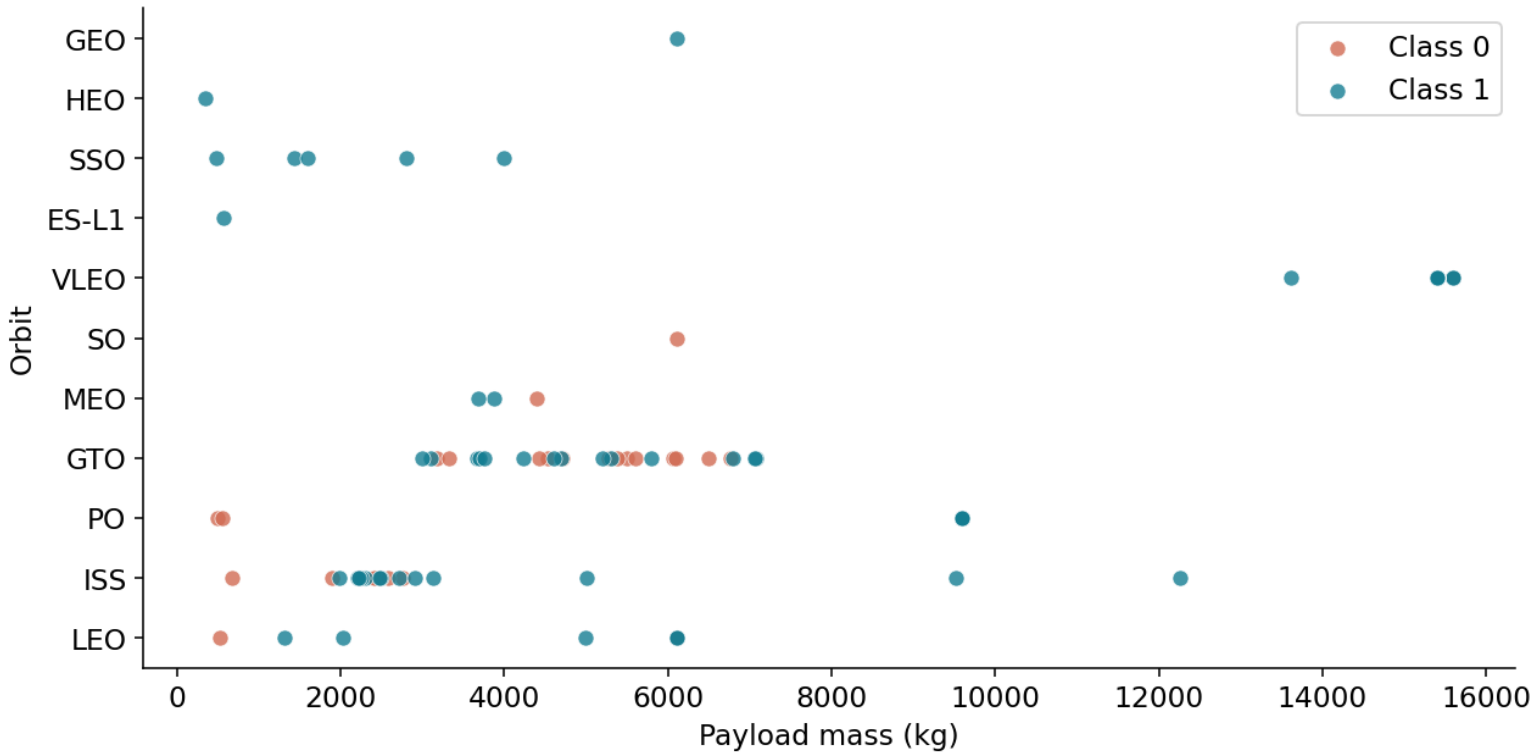
# Flight Number vs. Orbit Type




## Insights

- LEO success appears related to flight number
- GTO shows no clear link with flight number
- VLEO launches occur late and are mostly successful
- Later flights favour success across orbits

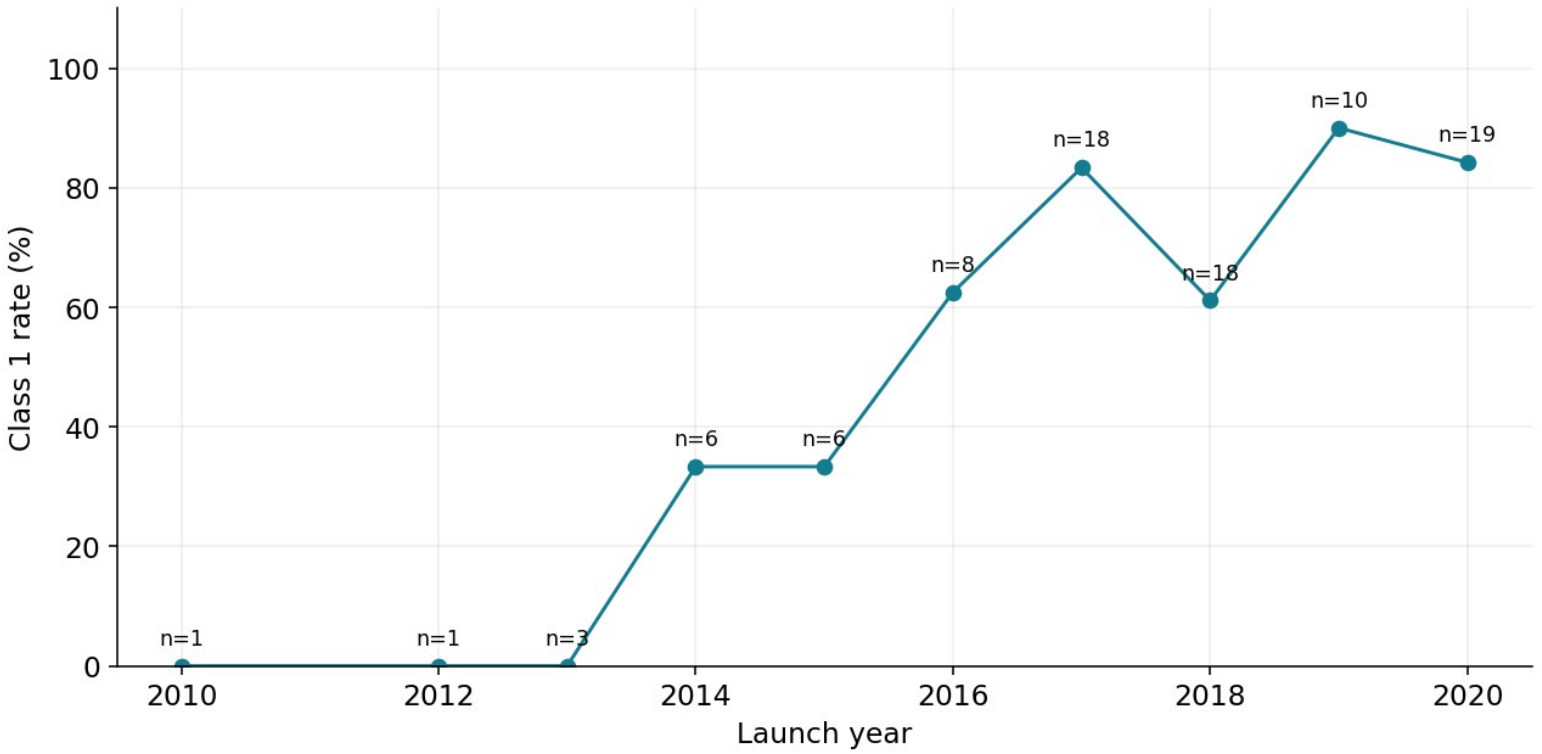
# Payload vs. Orbit Type




## Insights

- Heavy payloads succeed more for PO, LEO and ISS
- GTO mixes successes and failures across 3–7 t
- VLEO payloads cluster ~15 t and mostly land
- Orbit and payload interact

# Launch Success Yearly Trend




## Insights

- 0% success 2010–2013
- Steady rise from 2013, reaching 83.3% in 2017
- Dip to 61.1% in 2018, then 90% in 2019
- 84.2% in 2020 (n=19)

# All Launch Site Names

```
SELECT DISTINCT Launch_Site FROM launches ORDER BY Launch_Site;
```

| Launch_Site |
| --- |
| CCAFS LC-40 |
| CCAFS SLC-40 |
| KSC LC-39A |
| VAFB SLC-4E |

### Explanation

- DISTINCT returns each site once
- 4 unique launch sites: 3 in Florida, 1 in California
- CCAFS LC-40 is the older name of SLC-40

GitHub: https://github.com/YOUR-USERNAME/spacex-falcon9-capstone/blob/main/queries.sql  (also: analyze.py)

# Launch Site Names Beginning with 'CCA'

```
SELECT Date, Booster_Version, Launch_Site FROM launches
WHERE Launch_Site LIKE 'CCA%' ORDER BY Date LIMIT 5;
```

| Date | Booster_Version | Launch_Site |
|---|---|---|
| 2010-06-04 | F9 v1.0  B0003 | CCAFS LC-40 |
| 2010-12-08 | F9 v1.0  B0004 | CCAFS LC-40 |
| 2012-05-22 | F9 v1.0  B0005 | CCAFS LC-40 |
| 2012-10-08 | F9 v1.0  B0006 | CCAFS LC-40 |
| 2013-03-01 | F9 v1.0  B0007 | CCAFS LC-40 |

## Explanation

- LIKE 'CCA%' matches Cape Canaveral sites
- LIMIT 5 returns the first five
- All early launches were F9 v1.0 from CCAFS LC-40

GitHub: https://github.com/YOUR-USERNAME/spacex-falcon9-capstone/blob/main/queries.sql  (also: analyze.py)

# Total Payload Mass (NASA)

```
SELECT SUM(PAYLOAD_MASS__KG_) FROM launches WHERE Customer = 'NASA (CRS)';
SELECT SUM(PAYLOAD_MASS__KG_) FROM launches WHERE Customer LIKE '%NASA%';
```

| Customer filter | Total payload (kg) |
|---|---|
| NASA (CRS) | 45,596 |
| Any NASA customer | 107,010 |

## Explanation

- SUM() adds payload mass
- NASA (CRS) boosters carried 45,596 kg
- Including all NASA-related customers: 107,010 kg

GitHub: https://github.com/YOUR-USERNAME/spacex-falcon9-capstone/blob/main/queries.sql (also: analyze.py)

# Average Payload Mass by F9 v1.1

```
SELECT AVG(PAYLOAD_MASS__KG_) FROM launches WHERE Booster_Version LIKE 'F9 v1.1%';
```

| Booster | Average payload (kg) |
|---|---|
| F9 v1.1 (all variants) | 2,534.7 |

### Explanation

- AVG() over all F9 v1.1 boosters
- Average ≈ 2,535 kg
- Much lighter than later Block 5 loads (15,600 kg)

GitHub: https://github.com/YOUR-USERNAME/spacex-falcon9-capstone/blob/main/queries.sql  (also: analyze.py)

# First Successful Ground Landing Date

```
SELECT MIN(Date) FROM launches WHERE Landing_Outcome = 'Success (ground pad)';
```

| First successful ground-pad landing |
| --- |
| 2015-12-22 |

### Explanation

- MIN() returns earliest date
- First ground-pad landing: 22 December 2015
- Milestone that started reliable reuse

GitHub: https://github.com/YOUR-USERNAME/spacex-falcon9-capstone/blob/main/queries.sql  (also: analyze.py)

# Successful Drone Ship Landing with Payload 4,000–6,000 kg

```
SELECT Booster_Version, PAYLOAD_MASS__KG_ FROM launches WHERE Landing_Outcome='Success (drone ship)'
  AND PAYLOAD_MASS__KG_ > 4000 AND PAYLOAD_MASS__KG_ < 6000;
```

| Booster_Version | Payload (kg) |
|---|---|
| F9 FT B1022 | 4,696 |
| F9 FT B1026 | 4,600 |
| F9 FT  B1021.2 | 5,300 |
| F9 FT  B1031.2 | 5,200 |

## Explanation

- WHERE combines landing type and payload range
- 4 boosters, all F9 FT
- Two were reused boosters (.2)

GitHub: https://github.com/YOUR-USERNAME/spacex-falcon9-capstone/blob/main/queries.sql  (also: analyze.py)

# Total Number of Successful and Failure Mission Outcomes

```
SELECT Mission_Outcome, COUNT(*) FROM launches GROUP BY Mission_Outcome;
```

| Mission_Outcome | Count |
|---|---|
| Failure (in flight) | 1 |
| Success | 99 |
| Success (payload status unclear) | 1 |

### Explanation

- GROUP BY counts each outcome
- 100 successful missions (99 + 1 unclear payload status)
- Only 1 in-flight failure
- Mission success ≠ landing success

GitHub: https://github.com/YOUR-USERNAME/spacex-falcon9-capstone/blob/main/queries.sql  (also: analyze.py)

# Boosters Carried Maximum Payload

```
SELECT Booster_Version, PAYLOAD_MASS__KG_ FROM launches
WHERE PAYLOAD_MASS__KG_ = (SELECT MAX(PAYLOAD_MASS__KG_) FROM launches);
```

| Booster_Version | kg | Booster_Version | kg |
|---|---|---|---|
| F9 B5 B1048.4 | 15,600 | F9 B5 B1049.5 | 15,600 |
| F9 B5 B1049.4 | 15,600 | F9 B5 B1060.2 | 15,600 |
| F9 B5 B1051.3 | 15,600 | F9 B5 B1058.3 | 15,600 |
| F9 B5 B1056.4 | 15,600 | F9 B5 B1051.6 | 15,600 |
| F9 B5 B1048.5 | 15,600 | F9 B5 B1060.3 | 15,600 |
| F9 B5 B1051.4 | 15,600 | F9 B5 B1049.7 | 15,600 |

## Explanation

- Subquery finds the max payload
- 12 launches carried 15,600 kg
- All are Block 5 boosters, many reused several times

GitHub: https://github.com/YOUR-USERNAME/spacex-falcon9-capstone/blob/main/queries.sql (also: analyze.py)

# 2015 Launch Records

```
SELECT Date, Booster_Version, Launch_Site FROM launches
WHERE Date LIKE '2015%' AND Landing_Outcome = 'Failure (drone ship)';
```

| Date | Booster_Version | Launch_Site |
|---|---|---|
| 2015-01-10 | F9 v1.1 B1012 | CCAFS LC-40 |
| 2015-04-14 | F9 v1.1 B1015 | CCAFS LC-40 |

### Explanation

- Two failed drone-ship landings in 2015
- Both F9 v1.1 from CCAFS LC-40 (Jan & Apr)
- Early drone-ship attempts were difficult

GitHub: https://github.com/YOUR-USERNAME/spacex-falcon9-capstone/blob/main/queries.sql  (also: analyze.py)

# Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

```
SELECT Landing_Outcome, COUNT(*) AS count FROM launches WHERE Date BETWEEN '2010-06-04' AND '2017-03-20'
GROUP BY Landing_Outcome ORDER BY count DESC;
```

| Landing_Outcome | Count |
|---|---|
| No attempt | 10 |
| Failure (drone ship) | 5 |
| Success (drone ship) | 5 |
| Controlled (ocean) | 3 |
| Success (ground pad) | 3 |
| Failure (parachute) | 2 |
| Uncontrolled (ocean) | 2 |
| Precluded (drone ship) | 1 |

### Explanation

- 'No attempt' most common (10)
- Drone ship: 5 successes, 5 failures
- Only 3 ground-pad successes in this period
- Shows the early learning curve

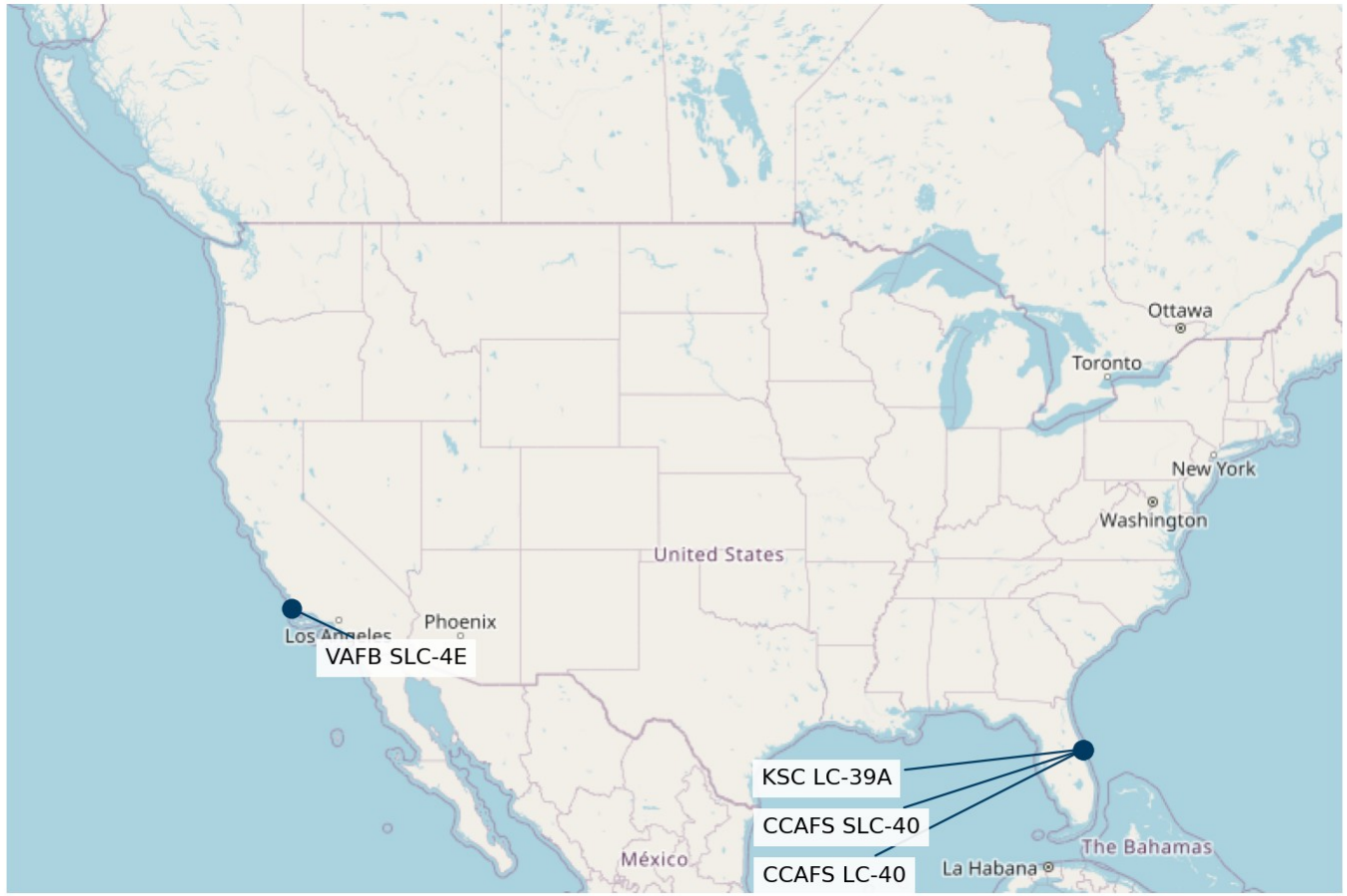
GitHub: https://github.com/YOUR-USERNAME/spacex-falcon9-capstone/blob/main/queries.sql  (also: analyze.py)

Section 3

# Launch Sites Proximity Analysis

Interactive Folium maps of launch sites, outcomes and surroundings

# All Launch Sites on a Map

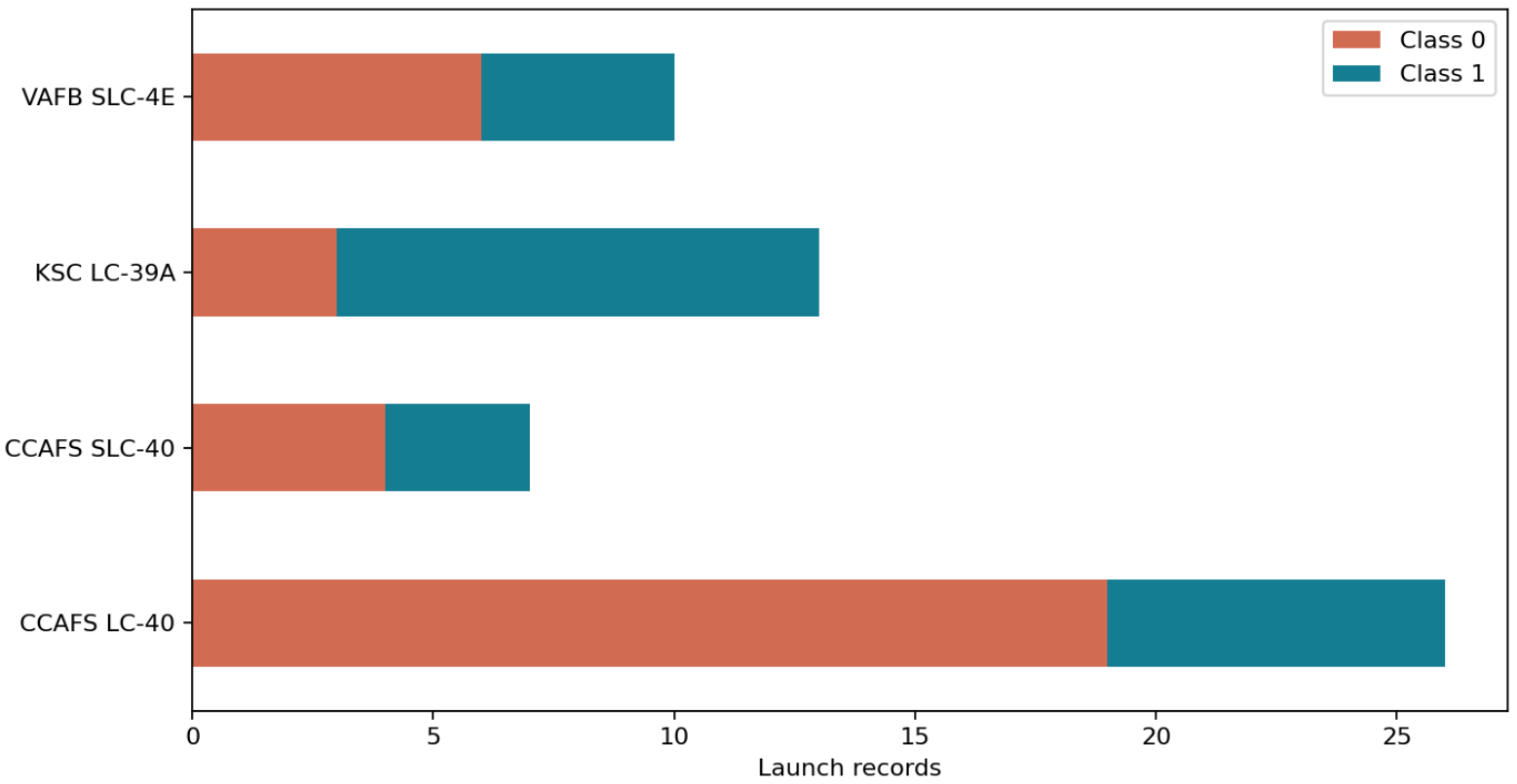



© OpenStreetMap contributors. Markers use IBM course coordinates.

## Findings

- VAFB SLC-4E on the California coast
- 3 sites at Cape Canaveral, Florida
- All sites are on the coast
- Florida sites are close to the equator for extra rotational speed

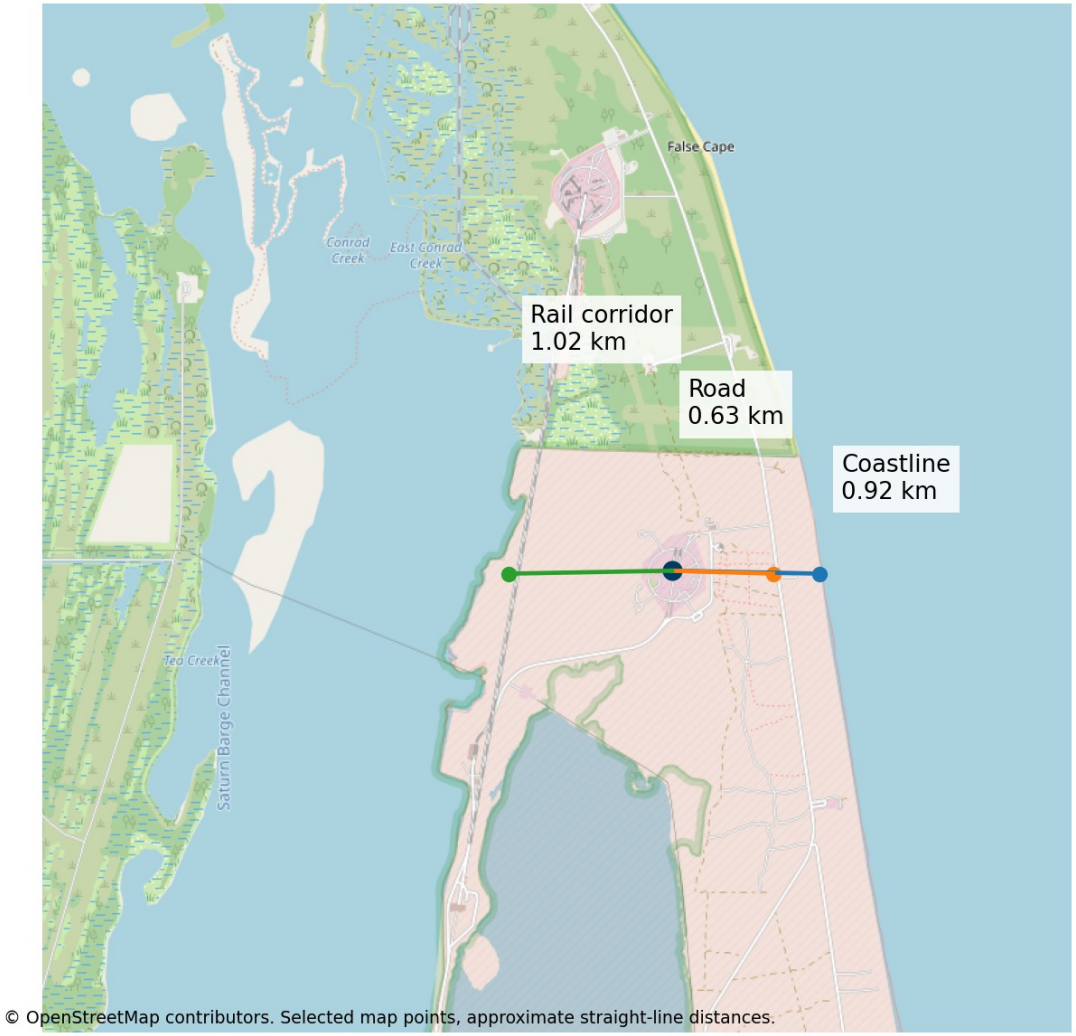
# Launch Outcomes by Site (Success vs. Failure Markers)




## Findings

- Folium marker clusters: green = success, red = failure; counts summarised here (teal = Class 1, orange = Class 0)
- KSC LC-39A: 10 of 13 successful (76.9%)
- CCAFS LC-40: mostly failures (7 of 26)
- Dashboard dataset: 56 launches

# Launch Site Proximity to Coastline, Road and Railway




| Feature (from CCAFS SLC-40) | Distance (km) |
|---|---|
| Road | 0.63 |
| Coastline | 0.92 |
| Rail corridor | 1.02 |

## Findings

- Sites are near coast for safe over-water trajectories
- Close to roads and rail for logistics
- Far from cities for safety
- Straight-line haversine distances

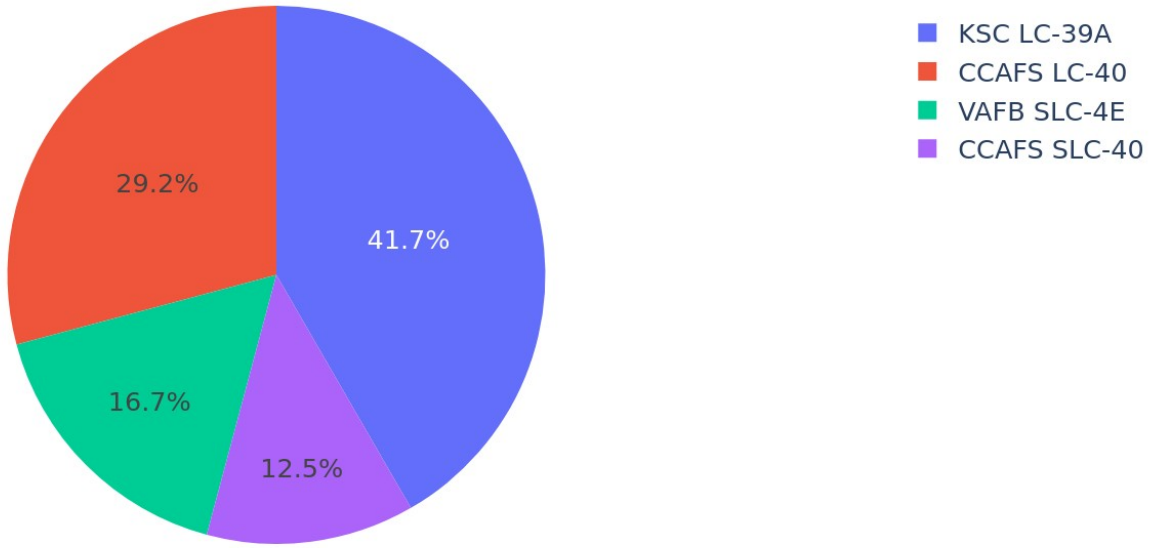
Section 4

# Build a Dashboard with Plotly Dash

Interactive success and payload analysis

# Launch Success Count for All Sites

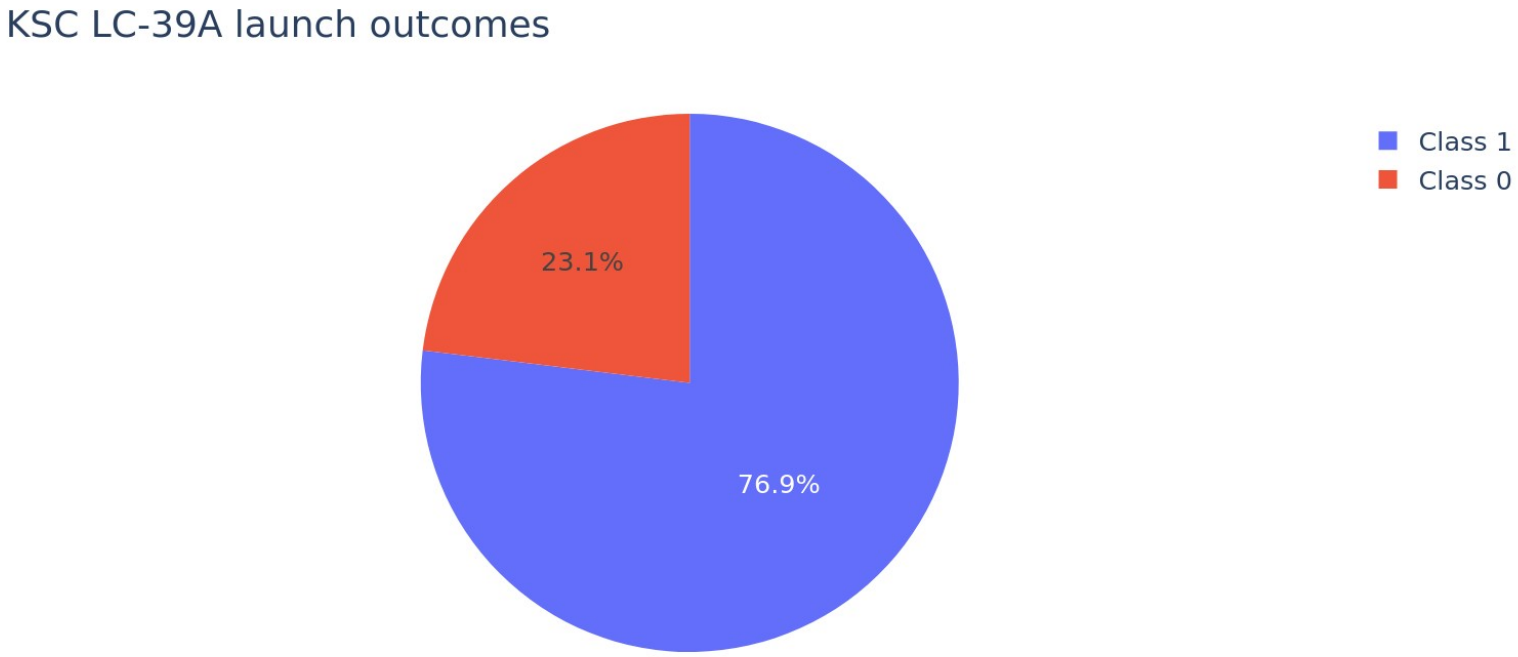
Successful launches by site (56-launch course snapshot)




## Findings

- KSC LC-39A has the largest share of successful launches (41.7%)
- CCAFS LC-40 next (29.2%)
- CCAFS SLC-40 lowest (12.5%)

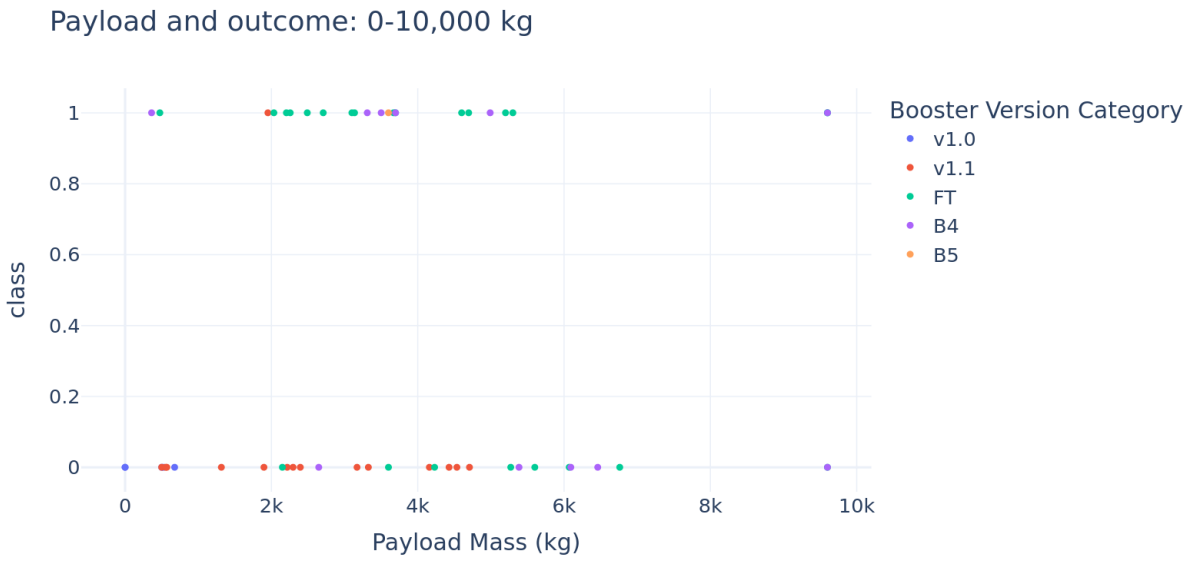
# Site with Highest Launch Success Ratio — KSC LC-39A



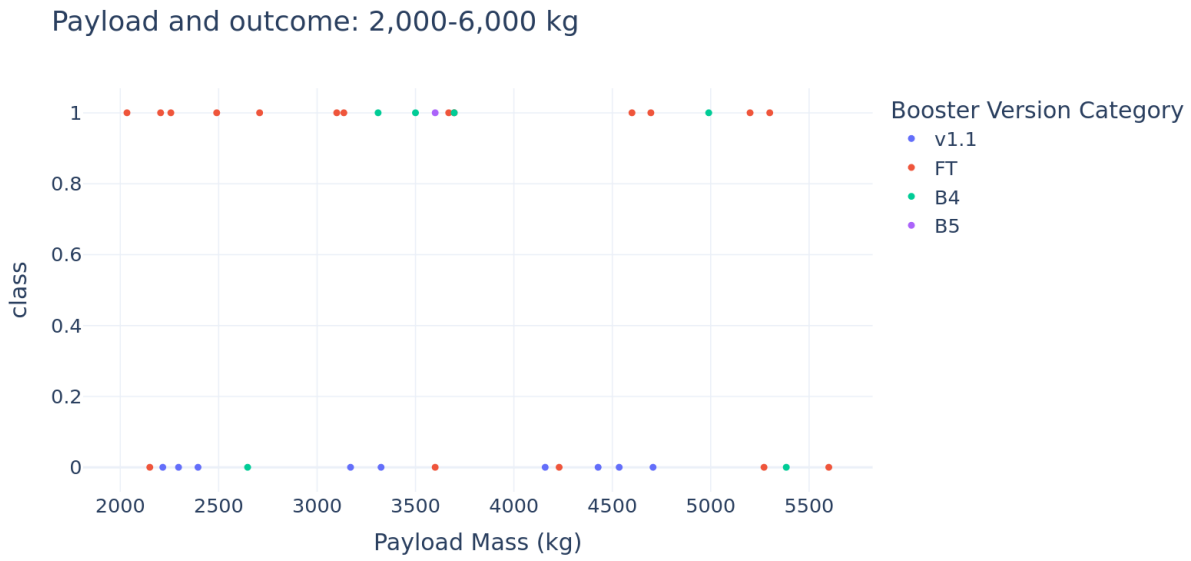

## Findings

- 76.9% success (10 of 13)
- 23.1% failures (3 of 13)
- Highest success ratio of any site

# Payload vs. Launch Outcome for All Sites




Full payload range 0–10,000 kg




Slider set to 2,000–6,000 kg

## Findings

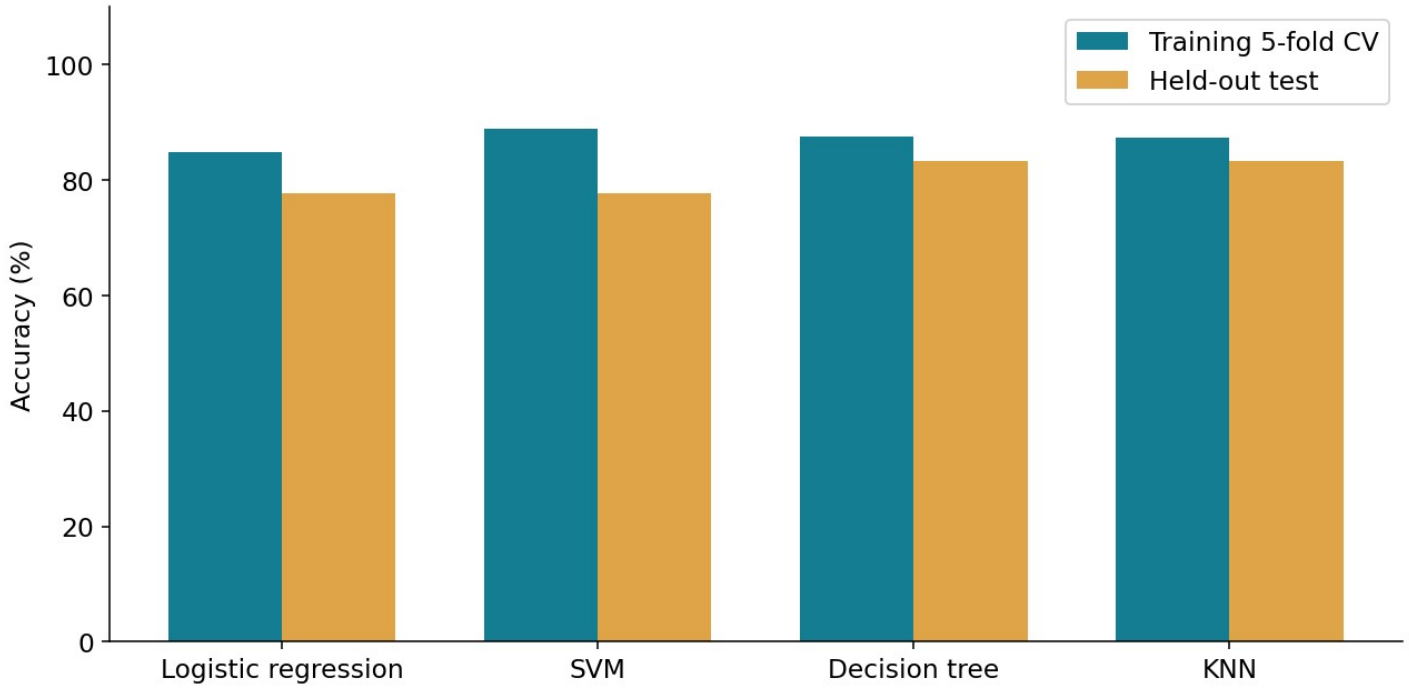
- FT boosters have the highest success rate; v1.1 boosters mostly failed
- Payloads of about 2,000–5,500 kg contain the most successful launches

Section 5

# Predictive Analysis (Classification)

Model comparison and evaluation

# Classification Accuracy

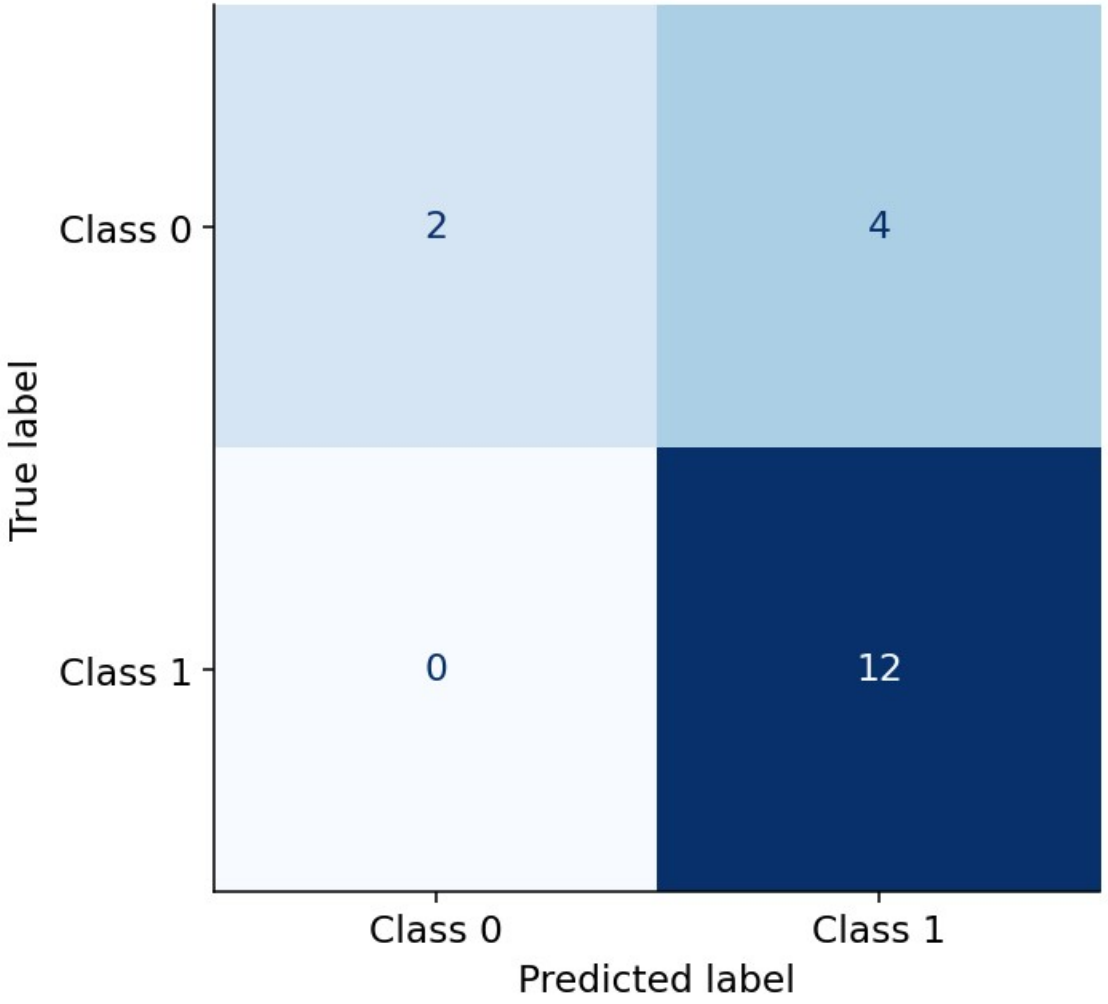



| Model | CV acc. | Test acc. | F1 |
|---|---|---|---|
| Logistic regression | 84.8% | 77.8% | 0.86 |
| SVM | 88.9% | 77.8% | 0.86 |
| Decision tree | 87.5% | 83.3% | 0.88 |
| KNN | 87.4% | 83.3% | 0.89 |

## Best model

- **SVM** (RBF, C=1) had the highest CV accuracy: 88.9%
- Decision tree & KNN scored highest on the test set (83.3%)
- 18-row test set: one prediction = 5.6 points

# Confusion Matrix — SVM (best CV model)




## Explanation

- 12 true positives: all actual landings were predicted correctly
- 0 false negatives
- 4 false positives: failures predicted as landings
- 2 true negatives
- The model is optimistic; it over-predicts landing
- Time-ordered check (last 18 flights): 88.9% vs 83.3% majority baseline

# Conclusions

- Landing success increases with flight number and over time: 0% in 2010–13 → 90% in 2019
- KSC LC-39A has the highest site success rate (76.9% in dashboard data, 77.3% in modeling data)
- SSO, GEO, HEO and ES-L1 orbits had 100% success; GTO was the weakest large orbit group (51.9%)
- FT and later boosters land far more often than v1.0/v1.1
- All launch sites are near the coast, roads and rail, but away from cities
- SVM achieved the best CV accuracy (88.9%); tree-based and KNN models generalised equally well on test (83.3%)
- **Business value:** the model can estimate landing probability, and therefore launch cost, before a launch, to support Space Y bids

# Creative and Innovative Insights

### What the data tells us beyond accuracy

- Model is never wrong on actual landings; it errs by being too optimistic — a costly error for a bidder
- Small test sets (18 rows) make accuracy coarse; CV and a time-ordered check give a more reliable view
- Some orbits have only 1 launch — 100% rates there are not reliable evidence

### Recommendations

- Use balanced accuracy and class weights to reduce false positives
- Add booster reuse history and weather data as features
- Collect newer launches (2021+) once the API is reachable
- Report predicted probability, not just a yes/no label, for cost estimation

# Appendix

- Project repository: https://github.com/YOUR-USERNAME/spacex-falcon9-capstone
- Files: SpaceX_Capstone.ipynb, api_collection.py, analyze.py, visuals.py, proximity.py, dashboard.py, queries.sql, results.json
- Datasets: dataset_part_1.csv (90 × 17, raw), dataset_part_2.csv (90, labelled), Spacex.csv (101 rows, SQL), spacex_launch_dash.csv (56 rows, dashboard)
- Different IBM datasets have different populations, so results are reported separately per dataset
- Map and proximity images use © OpenStreetMap contributors

